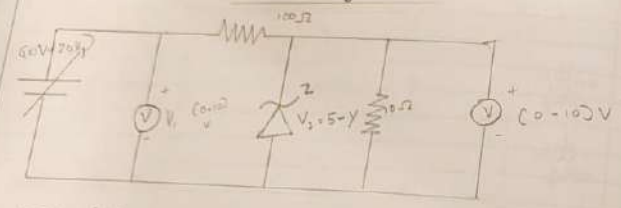
Circuit Diagram :-

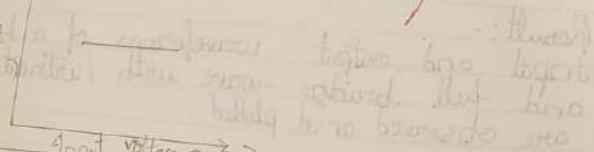
Line Regulation



Tabulation :-

S No	Input voltage V_1 (V)	Output voltage V_0 (V)
1	+12.0	+5.14
2	+13.0	+5.16
3	+14.0	+5.17
4	+15.0	+5.17
5	+16.0	+5.18
6	+17.0	+5.19

Model Graph :-



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ZENER DIODE AS VOLTAGE REGULATION

Aim :-

To simulate the line and load regulation operation zener diode.

Apparatus required :-

Laptop with proteus software.

Theory :-

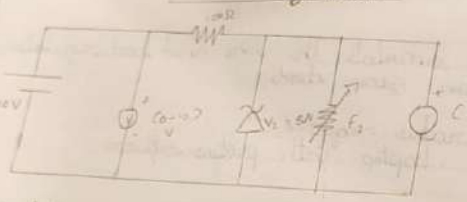
A heavily dopped PN junction diode is called zener diode. Due to heavily doped nature, zener diode works under forward bias and reverse bias conditions.

Zener diode under reverse bias condition is used as a voltage regulation. Even if the input voltage changes the output is constant. This is called line regulation.

If the load resistance changes, the output voltage is constant. This is called load regulation.

Circuit Diagram:-

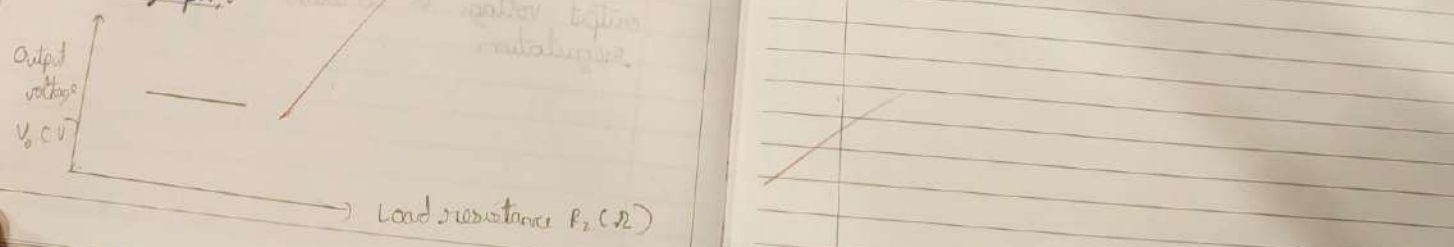
Load regulation



Tabulation :-

S No	Load resistance R_2 (Ω)	Output voltage V_o (V)
1	100Ω	+5.19
2	200Ω	+5.20
3	300Ω	+5.20
4	400Ω	+5.20
5	500Ω	+5.20
6	600Ω	+5.21

Model Graph:-



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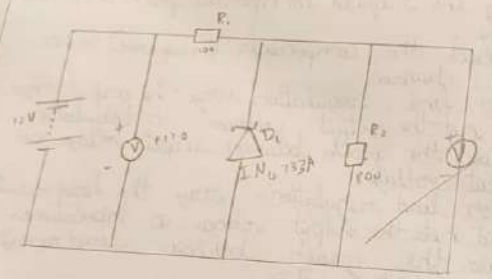
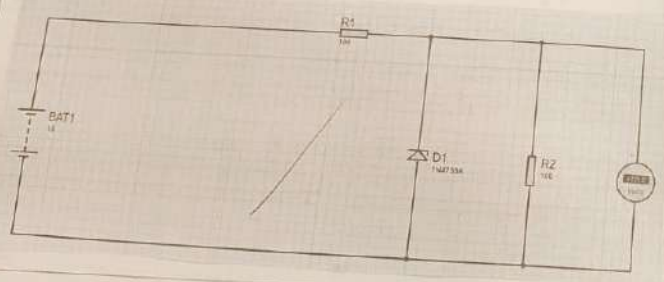
Procedure :-

1. Drag the required component from the proteus library.
2. Connect the components (required) as per circuit diagram.
3. For line regulation vary the input voltage and note the output voltages in tabulation. Draw the graph between input voltage and output voltage.
4. For load regulations vary the load resistance and note the output voltage in tabulations. Draw the graph between load resistance and output voltage.

Output :-





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Result :-

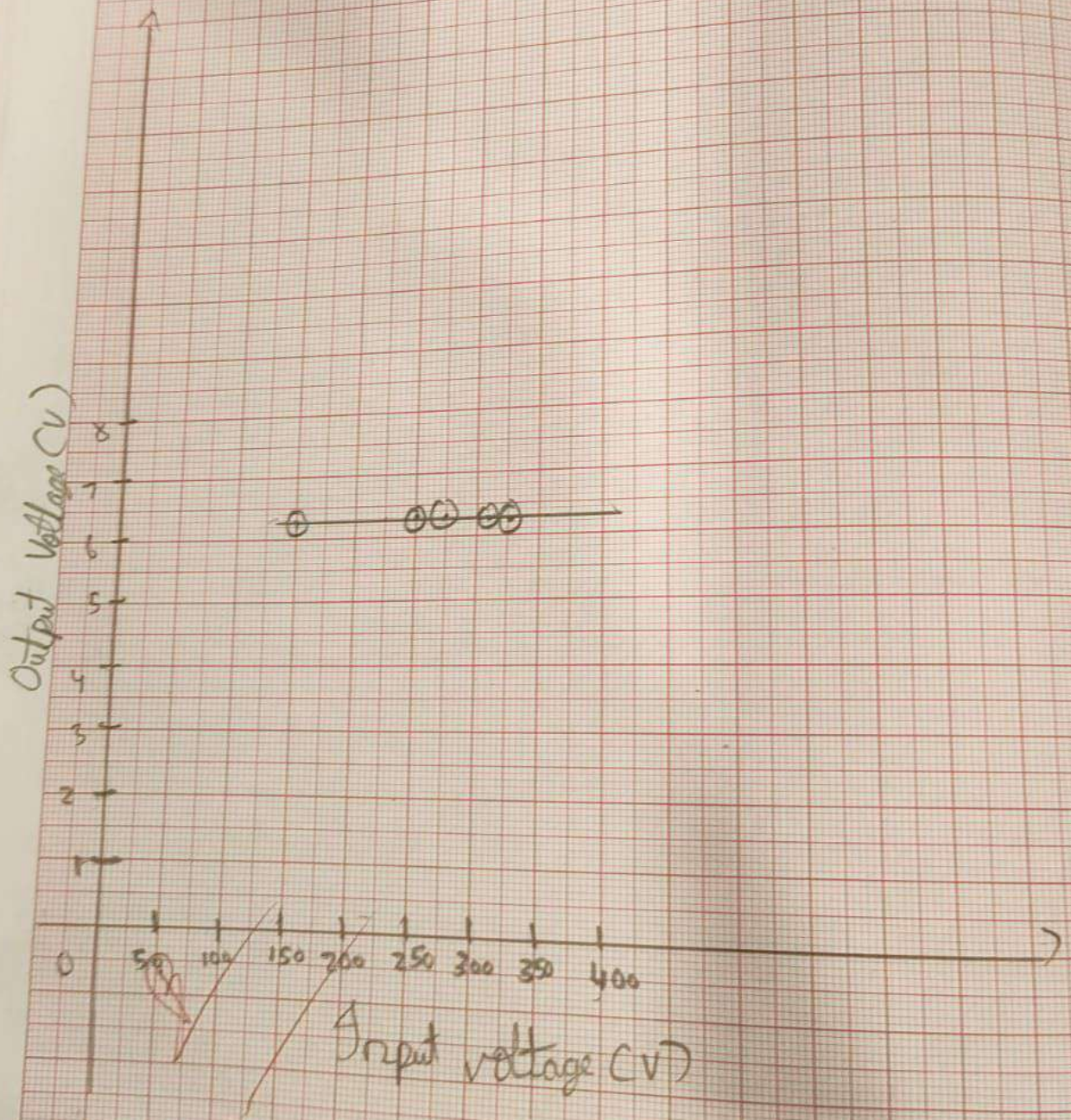
Thus simulation of line load regulation operation of zener diode is successfully completed.

Load regulation

Scale

Xaxis 1cm = 50 units

Yaxis 1cm = 1volt

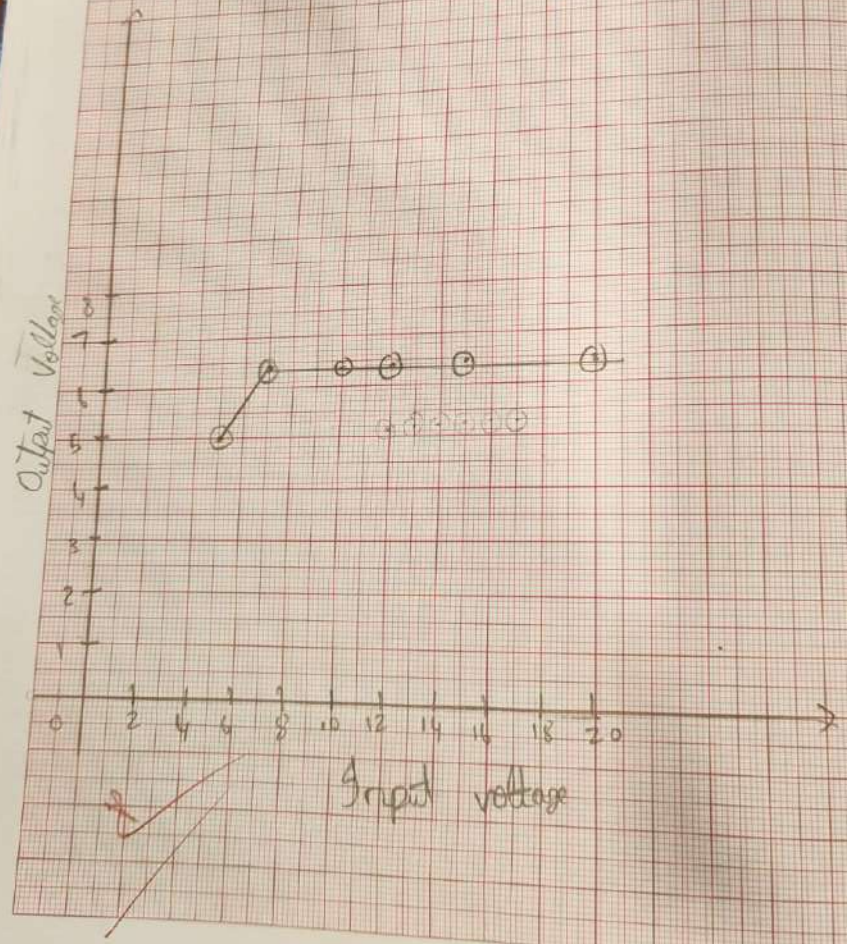


Line regulation

Scale

X axis 1cm = 2 volts

Y axis 1cm = 1 volt



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